
1. Introduction to the *Handbook of Innovation and Intellectual Property Rights*

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1. MOTIVATION OF THE *HANDBOOK*

Technological dynamism is central to social advancement, economic progress, and improved standards of living. Intellectual property rights (IPRs) are among the critical influences on innovation and creativity, along with other policies such as public funding, fiscal incentives, education, and training. They influence not only the generation of new knowledge but also the diffusion of and access to knowledge. This book project was initiated in the midst of the Covid-19 pandemic when gaining knowledge about the disease, combating it, and finding vaccines were a top global priority. It was during Covid that IP regimes and policies around the world came under more scrutiny as policymakers, scholars, and the media began debating whether the current systems in place adequately served their purpose of stimulating innovation and disseminating it in an efficient, timely, and equitable manner.

IPRs and IP regimes continue to be a dominant topic of conversation and research as artificial intelligence (AI) capabilities and usage have expanded in manifold directions. Issues such as the patenting and patentability of AI algorithms and machines, whether AI can be assigned IPRs, whether copyrights can be asserted over materials used in training datasets and the owners remunerated for them, the copyrightability of AI-generated works, and so forth, will need to be clarified, lest a persistent cloud of uncertainty hovers over AI developments and their applications.

The objective of this *Handbook* is to provide a compendium of some of the latest research on IP and innovation. The chapters survey and evaluate the state of the field, provide critical reflections, offer extensions to existing work, and help readers make sense of new research and the possible applications for it to modern challenges. I had envisioned this book to be useful to educators who may be teaching courses on innovation and IP and are looking for concise readings. Or to practitioners in government, the corporate world, the non-profit sector, or non-governmental organizations who are looking for critical summaries of particular topics. Or to students who want an overview of a topic before diving into their own original (and narrower) research. Or to our peers in academia or think tanks who either have not worked on innovation and IP for a while, or have worked on different aspects of innovation and IP, and so want to get up to speed on a topic. Or lastly to journalists and other professionals who want to be better informed. It is this vision that I shared with the authors who contributed to this book.

There are two value-added features of this book that I wish to mention. The first is the diversity of IPRs instruments that are investigated. In most previous books or survey articles, patents are the most often analyzed instrument of IPRs. While patents are important to study and are likely the most directly related to inventions per se, there is a rich panoply of other instruments of IPRs to consider, such as trade secrecy, copyrights, plant variety protection,

and trademarks, not to mention alternative approaches to innovation, such as open innovation, collective invention, or collectively owned IP, like traditional knowledge.

The second is the diversity of perspectives in this book. The contributors represent a spectrum of disciplines, such as law, economics, international business, management, and media studies. They provide country-specific as well as international perspectives, and sectoral as well as macro perspectives. The authors exploit a variety of approaches, and quantitative and/or qualitative methods of analysis. What the chapters all share in common are well-thought-out analyses by authors willing to push the boundaries to obtain a clearer understanding of an issue.

2. BACKGROUND ON IPRS

As a prelude to the chapter readings, I shall offer some background and context for how the field has evolved.¹ IP laws are national in scope, but therein is the reason I believe that IP issues have entered the international stage. While laws are national, markets are global. Knowledge has no natural boundary. It traverses or spills over across individuals and firms. It has the characteristics of a public good. Creators of IP in one market may have their creations copied or imitated by agents in another market where IP protection is absent. An international externality results whereby users in one jurisdiction can free-ride off the work of a creator in another jurisdiction. This negative externality may impair innovation and creativity if the free-riders succeed in exporting copies around the world (including possibly the creator's own market) and thereby dissipate the rewards to the creator.

The key governing international agreement on IPRs is the *Trade-Related Aspects of Intellectual Property Rights (TRIPS) Agreement* introduced in 1995 at the conclusion of the Uruguay Round negotiations (1986–1994). The agreement is administered by the World Trade Organization (WTO). Though IP laws remain national, member states have adopted the minimum standards required of the TRIPS agreement, resulting in some convergence of provisions across member jurisdictions. The agreement, however, is not commonly viewed as having been equally negotiated among member states. It largely reflects the systems that prevailed in developed countries, such as in the U.S. and Europe, which own most of the world's intellectual properties and had the most to gain from preventing copies. Developing countries therefore had the burden of making the most adjustments to their national laws. And there is still much debate about whether these “reforms” were conducive to economic development. Among other points, critics argue that stronger IP standards inhibit industrial development in poorer countries by restricting access to technology or limiting their ability to imitate or adapt technologies from abroad. Furthermore, since technology owners are largely in developed countries, the price increases of IP-related goods redistribute wealth from poor to rich nations.

Proponents of IP reform, on the other hand, argue that stronger IPRs should increase technology transfers to the developing world, and stimulate local innovation, as well as spur innovation in the developed world as the market for their innovation expands owing to reduced piracy, infringement, or imitation abroad. Proponents might also point to a provision in TRIPS (Article 66.2) that does enshrine a benefit for poor countries; namely, the article obliges developed countries to provide incentives for enterprises and institutions in their territories to help promote technology transfer to least-developed members “in order to enable them to create

a sound and viable technological base.” However, in reality, developed economies have fallen rather short of this obligation (see Park, 2017).

For innovation and technology transfers, the consensus seems to be that IPRs do not promote those activities by themselves, but in *conjunction* with other *complementary factors*. For example, skilled workers, access to finance, equipment, infrastructure, and other public policies, regulations, and tax regimes have to be conducive to innovation. The market environment should also be competitive. Economies that seem to have done well with their IPRs reforms were those that were export-oriented (e.g., Asian economies) or became more so. Open, competitive markets help to mitigate the market power effects of IPRs.

IPRs should also take into consideration the ecology of innovation. Creators form a chain. Each one is both a creator and a user who builds on prior work. IP protection that shows a preferential bias toward current IP owners would dampen future creations. The current IP owners or incumbent firms may not feel any competitive pressure to innovate and stay ahead of their rivals. In this sense, the focus should be on the *optimal* level of IPRs, not simply on the strength of protection and enforcement.

A large swath of developing economies have still not caught up technologically to developed countries, despite strengthening their IP systems to TRIPS standards. Most of them remain as technology importers. Studies and experiences have yet to demonstrate a strong connection between IPRs and measures of innovation in low- and middle-income economies. This could be due to a lag – that it takes more time for institutional changes to lead to increases in innovative capacity – or that the complementary factors are not yet present, such as a strong investment climate or adequate levels of human capital.

Another way to interpret these thoughts is that IPRs may have a significant impact on innovation after economies reach some threshold level of technological capabilities or absorptive capacities, which are the capacities to assimilate, absorb, and exploit valuable knowledge. Laws and regulations do not alter these capacities overnight. At best, they might create an environment conducive to business investment and human capital accumulation, but not build the underlying capabilities.

Then aligning IP systems appropriately to technological capabilities and other readiness factors would mean that the harmonization of IP laws and institutions globally is not optimal. IP standards should be differentiated by the level of economic development, which in turn depends on the maturity of industries, the industrial composition (whether agricultural, manufacturing, or mining), and the nature of technological competencies.

IP laws cover not only the rights of innovators and creators but also the rights of *users*. This matters for the ecology of innovation, as alluded to above, since producers and consumers of knowledge are interdependent. Laws can explicitly state what users are entitled to, or circumscribe the rights of IP owners through provisions on limitations and exceptions. These allow users to access or employ creations or innovations without seeking permission, such as fair use or fair dealing in copyright law, or farmer’s privilege in plant variety protection laws which allow farmers to re-use protected seeds without needing to compensate the breeder who owns the right. In patent law, working requirements exist that require the patent holder to practice the invention at some point – or else be subject to compulsory licensing if the technology is inadequately exploited and the market is under-supplied.

Thus, another way to differentiate IP systems across countries at different stages of development is to empower more flexible limitations and exceptions in the developing world. This enables the latter to better access technology and reduce the costs of conducting innovation.

Moreover, the limitations and exceptions may themselves foster valuable industry, as when the user makes transformative works out of existing IP. All of these free uses might actually enhance the value of the original works or technologies.

The scholarly community has learned a lot more about how innovation works, the needs of innovators, and the role of IP since the TRIPS agreement came into effect. Our understanding is more nuanced and exhibits awareness that one system of innovation governance cannot possibly fit all industrial and economic circumstances. Our understanding will often be tested, particularly when IP policies meet new challenges, whether it be a global health crisis or an emerging world of AI. The chapters to follow take stock of scholarship thus far and help advance conversations further.

3. SUMMARIES OF CHAPTERS

I have grouped the chapters by topic area. However, not only are there different ways to cluster the chapters, but many chapters can belong to multiple topic areas. It is not an easy task to define themes for this rich collection of chapters and then find the best fit for each chapter. But I did attempt this grouping nonetheless in order to give readers some organizing way of following the chapters. That said, readers will find quite a bit of overlap in coverage among the different contributions.

In Part I, the chapters provide an international, macroeconomic perspective. In Chapter 2, Chu surveys the research on the impact of IP on research and development (R&D) and economic growth. Overall, it provides a useful conceptual framework for thinking about the macroeconomic effects. The chapter indicates that IP can involve not just one policy lever but different instruments, be they fees, duration, or scope of protection. At present, more compelling evidence is needed to show how the developing world benefits from the IP system. Global coordination ought to be practiced, instead of expecting the developing world simply to adopt what the industrialized world established.

IP has expanded into a global field of study. What was once the domain of a closed-economy (or national) analysis, IP is central in studies of international trade and finance. The chapter by Maskus provides an insightful and masterful tour of the research on IPRs and trade. Through the course of the chapter, we learn the challenges of conducting empirical research on IP, exports, and global innovation, over matters such as establishing causality and dealing with selection biases, particularly regarding exporters that tend to be more productive, have skilled workers, and invest relatively more in R&D. The chapter also examines the role of regional trade agreements and multinational companies, which the chapters that follow it explore in more depth.

Campi and Dueñas write about recent research on preferential trade agreements (PTAs). The PTAs enable member states to typically augment IP protection beyond that set by multilateral agreements, like TRIPS. Not too surprisingly, regional agreements are quicker to structure since fewer signatory countries are involved. This chapter discusses the trends and implications of these *TRIPS-Plus* provisions. It delves into the databases available for researchers interested in studying PTAs. It discusses methodological challenges of identifying the effects of PTAs on trade flows, such as model specification, endogeneity, and country heterogeneity, and the difficulties of separating out the effects of IP chapters from other chapters in the same agreement.

Berry's chapter focuses on multinational companies as the conduit for knowledge. Specifically, it examines how their knowledge-related activities abroad affect their domestic innovation. The chapter uses microdata on U.S. multinational companies and merges them with U.S. patenting data. The higher the value-added activities of companies abroad, the greater the domestic innovations of U.S. companies. What drives this result, among other factors, is that foreign subsidiaries can tap into local knowledge and access skilled labor pools and technologically advanced suppliers. The domestic innovation response tends to be higher for firms in industries that are lagging in the global economy – creating a kind of catch-up effect.

Much of the research in international IP has focused on IPRs. However, as mentioned earlier, laws also contain provisions pertaining to the interests of users. Patent laws, for example, contain provisions on limitations, copyright laws on fair use or fair dealing, and plant variety protection laws on research exemptions and farmer's privileges. Such flexibilities are central to creating balance in IP systems. Saggi's chapter addresses one of the most important of these, namely compulsory licensing. It analyzes the different rationales for compulsory licensing, how it works, and the economic principles and experiences to date. The chapter explores how a mere threat or announcement of compulsory licensing could have real effects on behavior and how it can be complementary to price controls. It is entirely possible that a system can provide compulsory licensing to improve consumer access to technologies *and* generate incentives for innovation.

In Part II, the chapters address novel areas of patent research. Klein and Şener discuss the patent puzzle, whereby filings have increased (owing to stronger patent rights) but with no concomitant increases in the innovation rate. This appears to be at odds with Schumpeterian models. First, Klein and Şener find that Schumpeter rather anticipated some of the problems posed by monopoly power and cautioned about the effects it would have on technology diffusion. Second, to resolve the puzzle, the authors develop a model with deliberate rent-protection activities, such as litigation and lobbying by incumbent firms, and trade secrecy. Secrecy may better protect higher-quality improvements. Thus, firms with smaller quality improvements may select into patenting, thereby giving rise to the correlation between more patenting and less innovative breakthroughs.

In the chapter by de Rassenfosse and Higham, we learn how patent offices could function more efficiently if their patent system could incorporate a distributed ledger technology (DLT). A DLT is a record-keeping system via a network. A blockchain is an example of a DLT, one that allows for decentralized transactions; that is, no central authority is operating the system. The authors argue that a DLT could help patent systems deal with long grant lags, high instances of patent invalidity, and litigation. Their proposal is intriguing and worth serious consideration.

The chapter by Prud'homme also addresses problems with patent systems, namely patent thickets and the potential anti-commons effect of patent rights. The latter arises when too many agents own rights, such that the IP to an innovation is fragmented. The chapter examines whether state financing for patents might alleviate the anti-commons problem. State financing for patents may enable newly emerging companies to navigate around patent thickets. It might give companies leverage in cross-licensing negotiations or in preempting litigation. The chapter discusses China's experience with state financing and some of the missteps it took.

Duan's chapter has relevance for the problem of patent invalidity. It discusses the idea of patent title insurance against the possibility that a patent is later found invalid. The insurance

would compensate the title holder for the loss of any value due to the invalidation of a patent. Yet, the market for patent title insurance is virtually non-existent. (Insurance for litigation does exist but not one that can cover administrative challenges to a patent.) Duan argues that such a market might benefit society by providing a second layer of review of a patent after its examination at IP offices. But Duan does delve into why it might be difficult to create a market. Insurance companies, for instance, would need strong prior art search capabilities.

Part III turns to culture and IP. Three of the chapters deal with copyrights and the fourth with patents. The chapter by Nordås explores the impact of AI on the creation and distribution of digital music and addresses the current debates about the copyright implications of AI. It provides a timely discussion of whether AI-generated music should be copyrightable and if so, who is eligible to own the copyright. It also asks whether the training data for AI are subject to copyrights or come under fair use. AI is bound to make cultural goods trade a larger, more visible concern.

Handke's chapter also addresses copyright issues in the context of cultural industries. It discusses the key characteristics of cultural industries and reviews the main themes in cultural economics. There are special industrial organizational issues associated with cultural and creative industries. The chapter also discusses novel technologies that have surfaced in the cultural economics sphere, such as generative AI, which may lead to labor savings but also the potential displacement of professional creators. Generative AI requires tracking – to see where input works come from in the training data and how rightsholders are to be compensated, if at all, if their works can be tracked.

Aufderheide provides a critical chapter on the role of copyrights in innovation and creativity, pointing out that business support for copyrights can be at times anti-competitive. Important checks and balances are available through fair use, but the general public is not always aware of fair use or does not always understand it. This includes professionals, such as documentary filmmakers. The chapter discusses the reasons for this, and the consequences. The lack of knowledge of filmmakers' options to utilize copyright limitations and exceptions can put them in legal jeopardy. Creativity is also harmed because filmmakers would avoid projects that require extensive copyright clearances; projects especially affected have been those that documented popular culture, history, and politics.

While the chapter of Harutyunyan switches to patent rights, it continues with the theme of culture. The aim here is to show that the level of IP protection varies across nations depending upon where they fall along the individualistic and collectivist spectrum. After controlling for various aspects of culture, the analysis finds that individualism is positively correlated with the strength of patent rights. The implication here is that innovation tends to flourish in societies that emphasize personal achievements relative to those of the group. Collectivist societies, in contrast, emphasize group harmony and conformity. But some exceptions exist, such as South Korea, which has largely a collectivist culture but adopted institutions featuring strong IP. Over time, the country was able to align the concept of IP within its underlying culture.

Part IV investigates alternative approaches and instruments of IP, which tend to receive relatively scant attention in the academic literature as compared to patents and copyrights. We begin with a chapter on open science and innovation. Gold's chapter discusses the prospects for how open science partnerships can foster innovation. Open science here is distinguished from proprietary science and includes open data and open access (to journals). In open science partnerships, research outputs are not patentable, but firms can develop proprietary products based on shared knowledge. A case study is presented of the Montreal Neurological Institute

Hospital, whose adoption of open science principles for neurodegenerative diseases could be observed across its laboratories. Open science initiatives can help extricate research from harmful strategic behavior that can arise under the privatization of knowledge.

Schultz's chapter is on trade secrecy. This is an important contribution that should go a long way toward satisfying the demands in the academic literature for more research and analysis of the topic. The chapter discusses how secure legal protection for trade secrecy can actually enhance knowledge sharing among firms. The chapter provides the rationale and mechanisms by which this can occur. Using a case study of Covid-19, it shows how sharing secrets about novel processes facilitated the manufacturing partners to collaborate on research and production. The chapter argues against the involuntary transfers of know-how.

Next, Dutfield and Suthersanen discuss traditional knowledge, which too is a very important topic that has been, until now, under-studied. The authors make a strong case for giving the topic more attention. The chapter deliberates on the possibility of developing a binding legal instrument for the protection of traditional knowledge at the global level. In so doing, the chapter provides an extensive discussion of parallels and differences with land ownership. It points out the need to be wary of imposing laws and institutions upon other cultures. Historical precedent exists of how property-based economic systems led to Indigenous people experiencing dispossession.

The final two chapters in this part deal with trademarks. The chapter by Qian, Coughlan, and Kamate deals with branding and trademarks. Using a novel dataset of trademark infringement lawsuits in the U.S., it tests how such lawsuits are reflected in the returns to plaintiff and defendant firms. A verdict in favor of a plaintiff is associated with a positive change in abnormal returns and a negative one for defendants. The chapter is useful for showing how news regarding trademark litigation is received in the financial markets. Intangible assets are indeed a significant component of a firm's valuation.

The chapter by Qian, Gong, and Chen studies trademarks and competition from counterfeits. It provides a theoretical analysis of how counterfeiting can affect the nature and direction of authentic product innovation. There are two key dimensions of quality affected by counterfeiting: search quality (appearance) and experiential quality (functionality). Depending upon whether counterfeits are deceptive and whether a firm enjoys a cost advantage, the firm producing the authentic brand will optimally adjust the level of search quality and/or experiential quality. Thus, this chapter shows how the brand's product differentiation strategy responds to competition from counterfeiters.

The theme of Part V is to emphasize the different sectors of innovation and the value of treating sectors case by case. Industries possess such heterogeneities as to not always allow for generalizations about innovation and IP: product life, ease of imitation, and barriers to entry, among others. The following special areas are considered: agriculture, the environment, and health.

The chapter by Campi surveys the IP literature on agriculture. There has been much innovation in this sector, post-industrialization, including mechanization, capital improvements, and the green revolution. Agriculture is critical to ensuring global food security. Campi shows that agricultural R&D has expanded worldwide. The growth in private R&D, in particular, has outpaced that of public R&D in areas such as agricultural biotechnology and plant breeding. Plant breeder rights are protected by a system of plant variety protection (PVP). The chapter discusses the evolution of PVP globally. Laws on PVP also typically contain *flexibilities* such as breeders' exemption, which enables breeders to use protected varieties for purposes

of innovating new varieties; or farmers' privilege, which enables them to save seeds from one season's crops for resowing in the next season. Such flexibilities are important given the reproducibility of plant varieties and the cumulative nature of innovation. Studies to date are not fully conclusive as to whether IP protection effectively stimulates agricultural research.

Smith's chapter examines the role of IP in genetically modified organisms (GMO). More than 90 percent of the global land area for GMO production is in five countries, including the U.S. and Brazil, with soybeans being the leading crop. The chapter describes the unique attributes of GMO – self-replication and diffusion by nature – and the challenges they pose for IP enforcement. It reviews the ways in which the policy regimes on GMO vary by country. It conducts an analysis of a farmer's choices to adopt GMO with a license to re-use, adopt with no license to re-use, or adopt a non-GMO crop. These choices will vary by the strength of GMO patents across countries. Trade disputes will be pronounced in cases where non-licensed GMO seeds diffuse across regions.

The next chapter, by Nordhaus, deals with environmental matters, climate change, and technological innovation. A key motif is that green innovation is affected by a double externality. The first is an environmental one: the social cost of an environmental bad differs from the private cost. The environmental bad is underpriced. The second is an innovation externality due to the public goods aspect of knowledge, such that the private returns to innovation are below the true social returns. This chapter then builds on the framework to analyze incentives for green innovation, evaluates the promising technologies, and develops a case for carbon pricing.

The next three chapters are health-related. The chapter by Acri provides a comprehensive analysis of biopharmaceutical innovation. It highlights how risky R&D is in this sector and why IPRs have been traditionally relevant for this sector. The chapter provides an inside look at the pharmaceutical industry as well as the *generic* drug industry and covers products like biologics and biosimilars. It addresses matters that have deeply concerned researchers and policymakers alike, such as the role of data exclusivity and market exclusivity. It provides clarification on the concept of "evergreening" innovation (versus what would typically be incremental innovation aimed merely at extending IP protection) and the data challenges of identifying and assessing such innovation.

The chapter by Montobbio, Pellegrino, and Sterzi focuses on the patenting events surrounding the development of the mRNA Covid-19 vaccine. The study illustrates the interaction among biotech firms, universities, and research centers. It documents patent litigation and infringement cases owing to the fragmentation of IPRs. Multiple firms held patents with overlapping claims. These IP cases raised transactions costs. Though universities were instrumental in vaccine development, there are concerns that they did not have much control over what happened during the commercialization process. For dealing with these challenges, the chapter discusses policy options, such as patent pledges and patent pools.

Kilic's chapter argues that the Covid-19 pandemic exposed several weaknesses with the TRIPS system, as far as access to pharmaceuticals is concerned. TRIPS focuses on short-term measures, such as relying on "generic drug imports" in cases where pharmaceutical drugs are in short supply. Instead, there is a need to focus on long-term solutions such as developing local manufacturing capacities. The pandemic revealed that generic suppliers, like India, could not sufficiently supply large markets; that is, satisfy the large demand for Covid vaccines. The chapter provides arguments in favor of the TRIPS waiver. (The WTO had temporarily waived patent protection for Covid-19 vaccines during the pandemic.) The chapter argues that TRIPS

flexibilities as preferred by the developed world and by the pharmaceutical industry were (and are) inadequate. No doubt the debate will continue. Proponents of the waiver argue that developing countries would obtain medicines quicker, while opponents argue that it would jeopardize innovation or weaken a country's leadership in the technology.²

Part VI turns to regional analyses. Sharma and Dhanora's chapter provides an expert account of the evolution of India's patent system. The chapter provides a compact survey of the literature regarding patent policies and their influence and effects in India. The chapter serves as a great resource and introduction for those interested in researching IP and economic development, using India as a country study. It examines the history of legislation, discusses data on R&D and patents, and reviews the academic literature on the impact of those legislative changes on innovation and technology transfer, and the extent to which patents affect industrial market power and firm performance. It provides a useful agenda for follow-on researchers.

The chapter by Temel provides an inside look at Turkey's IP history. It summarizes thirty years of an ambitious agenda to boost patents and their commercialization. In 2010, a Vision 2023 plan was initiated to turn Turkey into a top large economy. The strategies for this pursuit were built around developing university–industry collaborations. Universities played a leading role in erecting funding schemes, creating technology transfer offices (TTOs), and implementing an entrepreneurial curriculum. Turkey also spurred science parks and instituted a version of the Bayh–Dole Act of the United States – namely, legislation that allows universities that receive government funding to own inventions rather than assign ownership to the government. Temel discusses where Turkey fell short of its goals, and why. Among the lessons are that TTOs needed more financial and personnel support and improved governance, and that the country needed to prioritize commercialization in the domestic economy rather than in global markets.

The chapter by Dosso studies the role of IP and startup companies in sub-Saharan Africa. There is much innovation promise and emerging vibrancy in the region. The chapter fills a huge vacuum in that studies on African IP and innovation are vexingly sparse. Dosso provides evidence and extensive commentary on the startup and venture capital landscape, discusses the institutions and relevant legal Acts, and provides policy recommendations for strengthening and sustaining startup activity in the region, much of which concerns funding gaps. As IP activities in sub-Saharan Africa expand, more evidence should become available for empirical analyses and for deriving insights into the mechanisms for economic development in the region.

4. GOING FORWARD: IMPLICATIONS FOR AI

As noted earlier, one of the next major challenges for IP systems and regulations is AI and AI-driven innovations and AI-generated products. Already for AI, scholarship has emerged in the area of IP and innovation. For example, the volume of articles in Abbott (2022) addresses the legal aspects of IP and AI, while that in Agrawal et al. (2019) addresses the economic impacts.

The chapters in this *Handbook* should help provide the foundation for understanding the roles of IP and explaining the impacts of IP policies. While AI will be disruptive for IP systems and for economic activity, and thereby create a bewildering array of questions, it will prove to

be useful for researchers and practitioners to draw back on the basics, the roots of IP principles, and the theories and evidence thus far for purposes of analyzing the implications of AI.

The following is a non-exhaustive list of topics relevant to AI research that are covered in this *Handbook*:

- Copyright and training data
- Culture
- Patentability by machine and non-obviousness
- Secrecy
- Litigation
- Trademarks
- Open source/innovation
- Academic research
- Macroeconomic implications for productivity growth and investment
- Imitation risk
- A Global North–South divide and development gaps
- Trade and multinational corporations
- Knowledge access and knowledge spillovers
- Information databases and datasets
- Market concentration
- Sectors where AI could be deployed: health and medicines, agriculture, and the environment

Of these, I would be concerned and curious about how AI affects the global distribution of income. Currently, AI development is concentrated mostly in the developed world, conducted by leading companies and universities. AI training datasets are expensive and foundation models resource-intensive. It takes a lot of energy (electricity) to power the models. These are resources, skill sets, and capabilities the Global South does not presently have. Unless AI technologies and knowledge diffuse globally, and consumers and producers share in the benefits, the North–South gap could widen, a gap that IP advocates and critics alike were hoping to prevent or narrow under IP systems.

To prevent AI development from magnifying global inequalities (or even inequalities within countries), AI development should become more inclusive, involving developers and users from across the globe. IP policies should be designed to facilitate this (and de-emphasize global AI leadership races). If collaboration does not happen naturally via markets, intergovernmental coordination may be required to involve the less developed countries in research, learning, and accessing innovations. It should be of mutual interest to promote balanced economic development via technical assistance. Lack of it not only harms poorer countries, but results in lost potential markets and economic opportunities for developed countries as well.

NOTES

1. My own survey of the field from ten years ago can be found in Park (2014).
2. In this case, it would be U.S. leadership. See <https://phrma.org/policy-issues/International/TRIPS-Waiver>.

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